Installing Carpet

Hungrysoft Company is renting a new office. The office is a polygon in a 2D plane with a Cartesian coordinate system. The polygon has N *distinct* vertices, numbered from 0 to $N-1$. For each i such that $0 \leq i < N$, vertex i is located at $(X[i], Y[i])$, where $X[i]$ and $Y[i]$ are nonnegative integers. There is an edge connecting vertex i and vertex $i+1$ (for each i such that $0 \leq i \leq N-2$), as well as an edge connecting vertex $N-1$ and vertex 0.

The office is an **axis-aligned** polygon, which means that each of the edges is parallel to either the x-axis or the y-axis. Moreover, the office is a **simple** polygon, that is:

- exactly two edges meet at each vertex;
- any pair of edges can only meet at a vertex.

The company has infinitely many pieces of carpet. Each piece is a square of side 2. The company would like to cover a big part of the office with these pieces. Specifically, the company wants to pick some vertical line and cover the part of the office to the left of the line.

For any integer k, let L_k denote the vertical line consisting of points with x-coordinate equal to k. A **covering** of the part of the office to the left of L_k is a placement of some number of pieces in the plane such that:

- each point which lies in the interior of the polygon *and* has x-coordinate less than k is covered by some piece;
- no point which lies outside of the polygon *or* has x-coordinate greater than k is covered by some piece;
- the interiors of the pieces do not overlap.

The minimum x-coordinate of any vertex in the office is 0. Let M denote the maximum x-coordinate of any vertex in the office. Your task is to help Hungrysoft determine the largest integer k, such that $k \leq M$, and there exists a covering of the part of the office to the left of L_k. Note that by definition, there exists a covering of the part of the office to the left of L_0 (which uses 0 pieces).

Implementation Details

Do not read from standard input or print to standard output. Instead, you should implement the following procedure:

```
int max_line(int N, int M, int[] X, int[] Y)
```

- N: the number of vertices.
- M: the maximum x-coordinate of any vertex.
- X, Y: arrays of length N describing the coordinates of the vertices.
- This procedure should return the maximum k such that $k \leq M$ and there exists a covering of the part of the office to the left of L_k.
- This procedure is called exactly once for each test case.

Examples

Example 1

Consider the following call:

```
max_line(14, 6,
         [0, 0, 2, 2, 0, 0, 3, 3, 4, 6, 6, 3, 3, 3],
         [1, 3, 3, 4, 4, 6, 6, 7, 7, 7, 5, 5, 2, 1])
```

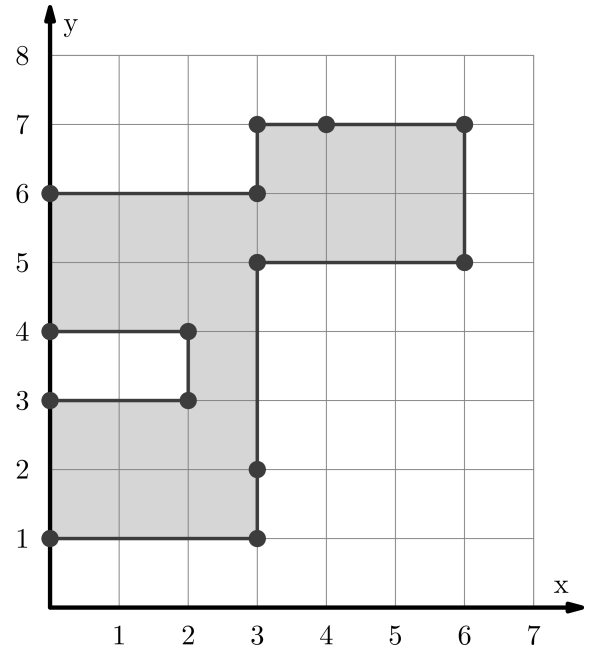
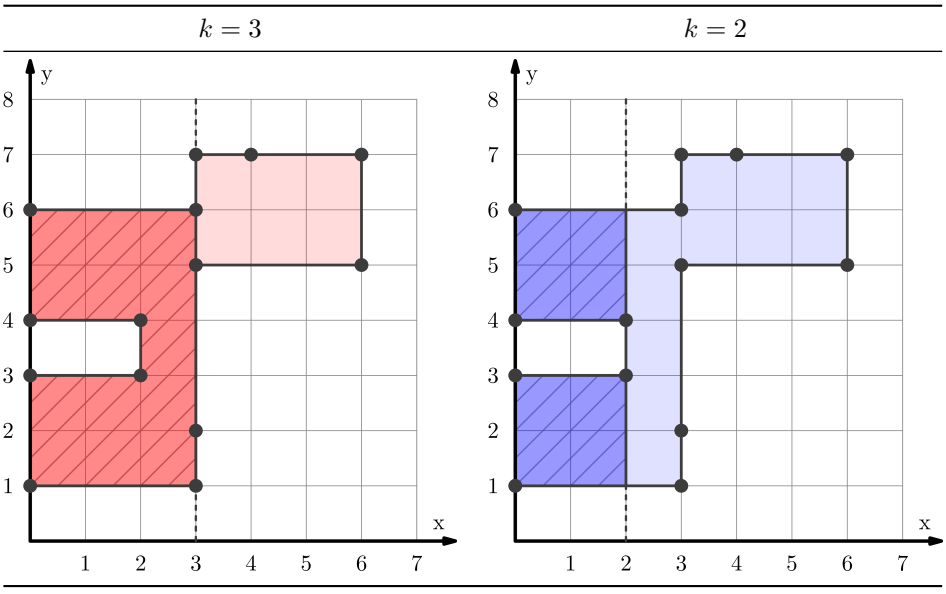


Figure 1: The office is marked as a shaded area in the picture above.

The following pictures show the part of the office to the left of line L_k for two different values of k.

- The left picture corresponds to $k = 3$. There is no covering of the part of the office to the left of L_3.
- The right picture corresponds to $k = 2$. There is a covering of the part of the office to the left of L_2. The covering uses two pieces.



For any $k > 2$ there is no covering of the part of the office to the left of L_k. Hence, the procedure should return 2.

Example 2

Consider the following call:

```
max_line(18, 9,
         [0, 2, 2, 4, 4, 9, 9, 4, 4, 7, 7, 9, 9, 4, 4, 2, 2, 0],
         [2, 2, 1, 1, 0, 0, 2, 2, 4, 4, 3, 3, 6, 6, 5, 5, 4, 4])
```

As illustrated below, it is possible to cover the part of the office to the left of line L_6.

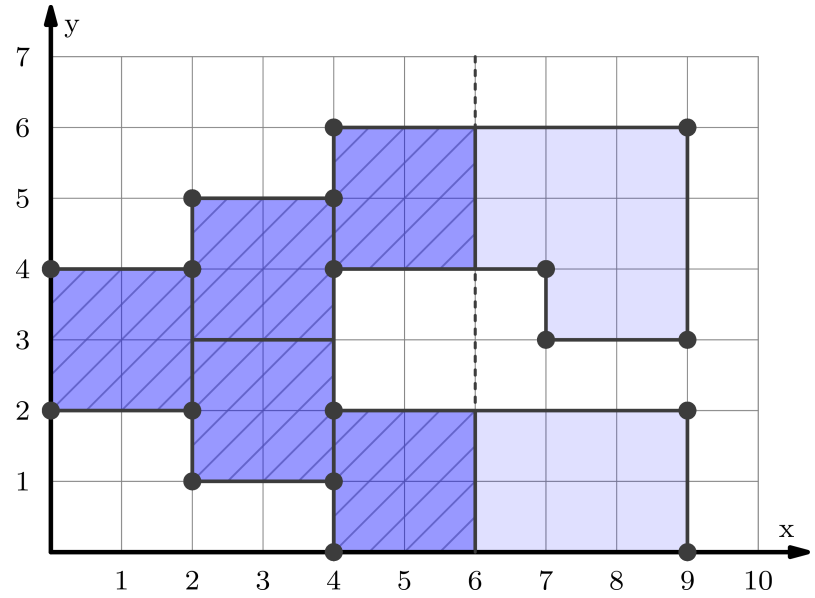


Figure 2: Covering the office to the left of line L_6

For each $k > 6$ there is no covering of the part of the office to the left of L_k. Hence, the procedure should return 6.

Example 3

Consider the following call:

```
max_line(4, 3, [0, 0, 3, 3], [0, 3, 3, 0])
```

There is no positive value of k such that it is possible to cover the part of the office to the left of L_k. Hence, the procedure should return 0.

Constraints

- $4 \leq N \leq 200\,000$
- $1 \leq M \leq 10^9$
- The minimum of $X[0], X[1], \ldots, X[N-1]$ is 0.
- The maximum of $X[0], X[1], \ldots, X[N-1]$ is M.
- $0 \leq Y[i] \leq 10^9$ (for each i such that $0 \leq i < N$)
- The office forms an axis-aligned simple polygon.

Subtasks

1. (5 points) $N = 4$
2. (11 points) $N \leq 6$
3. (12 points) $X[N-1] = 0$, $Y[N-1] = 0$, $X[i] \leq X[i+1]$, $Y[i] \geq Y[i+1]$ (for each i such that $0 \leq i \leq N-3$)
4. (18 points) $M \leq 1\,000$, $Y[i] \leq 1\,000$ (for each i such that $0 \leq i < N$)
5. (20 points) $Y[i]$ is even (for each i such that $0 \leq i < N$)
6. (24 points) $X[i]$ is even (for each i such that $0 \leq i < N$)
7. (10 points) No additional constraints.

Sample Grader

The sample grader reads the input in the following format:

- line 1: N M
- line $2 + i$ $(0 \leq i < N)$: $X[i]$ $Y[i]$

The sample grader prints your answer in the following format:

- line 1: the return value of `max_line`